GameFi Economy

ERC-1155 · Crafting · AMM · DAO · Rental Vault

Solidity ^0.8.25 · Foundry · Chainlink VRF v2.5 · OpenZeppelin v5 · The Graph

ERC-1155
ERC-4626
ERC-20Votes
CREATE2
UUPS
VRF v2.5
Price Feeds

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Protocol Overview

A fully on-chain gaming economy on Arbitrum Sepolia

ERC-1155 Economy

6 fungible elements + 2 semi-fungible pony NFTs in one Equestria1155 contract

Crafting Engine

Burn elements → VRF request → mint pony NFT (10% double-drop chance)

Constant-Product AMM

0.3% fee swap between ERC-20 game tokens; Yul assembly getAmountOutYul optimisation

DAO Governance

1M GOV token split 40/30/20/10%; MyGovernor + TimelockController; Treasury & Box

NFT Rental Vault

ERC-4626; NFT stakers get 120% share boost; Timelock injects yield

Chainlink

VRF v2.5 for randomness; Price Feeds with staleness/round checks; Arbitrum Sepolia

ERC-1155 Token Design

Equestria1155 — custom implementation with VRF crafting

ELEMENT TOKENS (IDs 1–6)

1	HONESTY
2	KINDNESS
3	LAUGHTER
4	GENEROSITY
5	LOYALTY
6	MAGIC

PONY NFTs (IDs 1001–1002)

ID 1001 · PINKIE_PIE

Recipe: 15 HON + 10 KIND + 100 LAUGH + 20 GEN + 5 LOY + 0 MAG

craft() burns elements → requestRandomWords → fulfillRandomWords mints pony
10% chance of double drop (randomWord % 100 < 10)

ID 1002 · STARLIGHT_GLIMMER

Recipe: 5 HON + 3 KIND + 0 LAUGH + 0 GEN + 15 LOY + 80 MAG

craft() burns elements → requestRandomWords → fulfillRandomWords mints pony
10% chance of double drop (randomWord % 100 < 10)

Core Smart Contracts

13 contracts across 5 modules — src/

Contract	Standard / Pattern	Responsibility
Equestria1155	ERC-1155 + VRFConsumerBaseV2Plus	Crafting engine; element + pony tokens; VRF callback
AMM	Constant-product AMM	0.3% fee; addLiquidity/removeLiquidity/swap; Yul optimisation
LPToken	ERC-20	AMM share; deployed by AMM constructor; mint/burn by AMM only
GameToken	ERC-20 + Ownable	Mintable game token; pair tokens HAR & SPK
NFTRentalVault	ERC-4626 + ReentrancyGuard	GOV as asset; NFT stakers get 120% boost; injectYield()
GovernanceToken	ERC-20Votes + Permit	1M GOV; 40/30/20/10% split at constructor
MyGovernor	OZ Governor (5 extensions)	4% quorum; 1% proposal threshold; Timelock execution
Treasury	Custom + Timelock ownership	ETH + ERC-20 withdrawals; onlyTimelock guard
TokenVesting	Linear vesting + SafeERC20	365-day schedule; release() permissionless
GameTokenFactory	CREATE + CREATE2	createToken2() deterministic; predictAddress()
GameTokenV1/V2	ERC-20Upgradeable + UUPS	Proxy upgrade demo; V2 adds burn(); onlyOwner authorize
PriceFeedConsumer	AggregatorV3Interface	Staleness + round + negative price guards
VRFConsumer	VRFConsumerBaseV2Plus	Standalone random; requestId→sender map

AMM & Token Factory

Constant-product market maker + CREATE2 factory

AMM.sol — Uniswap V2 style

Fee	0.3% (factor 997/1000 in getAmountOut)
Formula	amountOut = (amIn×997×rOut) / (rIn×1000 + amIn×997)
Yul	getAmountOutYul() — memory-safe assembly optimisation
LP Token	LPToken ERC-20; minted via √(amA×amB) on first deposit
Slippage	minAmountA/B on add; minAmountOut on swap
Safety	ReentrancyGuard; amountOut ≥ reserveOut reverts (no drain)
Pair	HAR (Harmony) + SPK (Sparkle) — both GameToken ERC-20
Quote	quoteAForB() / quoteBForA() view functions

GameTokenFactory.sol — CREATE / CREATE2

- → createToken() — unpredictable address (CREATE)
- → createToken2(salt) — deterministic address (CREATE2)
- → predictAddress(name, symbol, salt) — compute before deploy
- → allTokens[] tracks all deployed token addresses

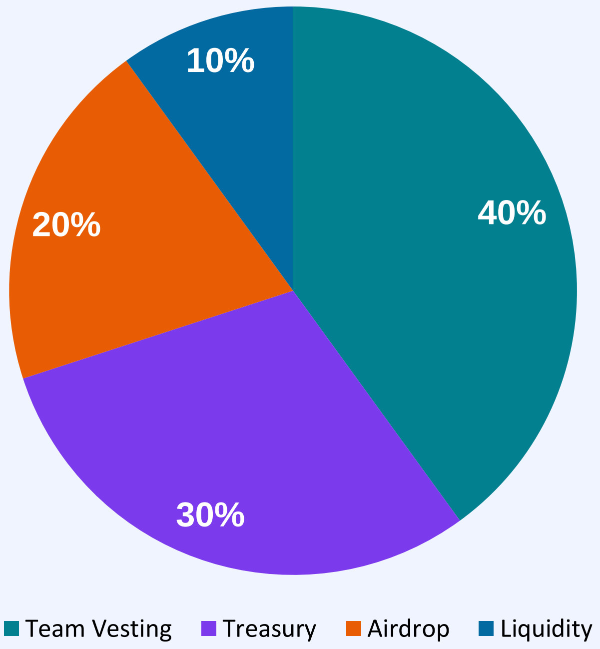
GameTokenV1/V2 — UUPS Upgradeable

- → _disableInitializers() in constructor
- → initialize(name, symbol, owner) — replaces constructor
- → _authorizeUpgrade() restricted to owner
- → V2 adds burn() — additive upgrade, no storage collision
- → ERC1967Proxy deployed in Deploy.s.sol

DAO Governance

GovernanceToken · MyGovernor · TimelockController · Treasury

1,000,000 GOV Token Distribution



MyGovernor Parameters

Quorum	4% of total supply
Proposal Threshold	1% of getPastTotalSupply
Voting Delay	5 blocks (testnet)
Voting Period	10 blocks (testnet)
Timelock Delay	10 s testnet / 2d+ production
Token	ERC-20Votes + ERC-20Permit
Governor Exts	Settings, Counting, Votes, QuorumFraction, Timelock

NFT Rental Vault

ERC-4626 vault with ERC-1155 staking boost

1. Deposit GOV Tokens → 2. Receive vShares → 3. Stake PINKIE PIE (id 1001) → 4. Get 120% Boost → 5. Yield Injected by Timelock

Asset & Shares

GovernanceToken (GOV) as vault asset; vSHARE ERC-20 for LP positions

NFT Gating

Only boostedNftId (PINKIE_PIE = 1001) accepted; AlreadyStaked guard; unstakeNFT() returns NFT to renter

IERC1155Receiver

onERC1155Received + onERC1155BatchReceived implemented; supportsInterface returns IERC1155Receiver.interfaceId

Boost Mechanism

_deposit() checks isRenting[receiver] → applies boostBps (default 1200/1000 = 120%). Configurable via setBoostBps() (owner only)

Yield Injection

injectYield(amount) pulls GOV from owner (Timelock) into vault, increasing asset/share ratio for all holders

Chainlink Integrations

VRF v2.5 · Price Feeds · Arbitrum Sepolia config

VRF v2.5 — Verifiable Randomness

Coordinator (Arb Sepolia):	0x5CE8D5A2BC84beb22a398CCA51996F7930313D61
Key Hash:	0x1770bdc7...f6cfd2be
REQUEST_CONFIRMATIONS:	3 blocks
CALLBACK_GAS_LIMIT (craft):	200,000 gas
CALLBACK_GAS_LIMIT (vrf):	100,000 gas
Num Words:	1 per request
fulfillRandomWords:	internal override; only VRF coordinator
Double-drop:	10% if randomWord % 100 < 10

Price Feed Consumer

ETH/USD (Arb Sepolia):

0xd30e2101a97dcbAeBCBC04F14C3f624E67A35165

Staleness Check

block.timestamp − updatedAt > threshold → revert StalePrice()

Invalid Round

startedAt == 0 → revert; answeredInRound < roundId → revert

Negative Price

price_ < 1 → revert InvalidPrice()

Threshold

10 min testnet; recommend 3600 s (1 hr) for mainnet ETH/USD

Deployment & Frontend

Deploy.s.sol · GitHub Actions CI · Next.js dApp · The Graph

1. **GovernanceToken**
   Mints 40/30/20/10% to team/treasury/airdrop/liquidity
2. **TokenVesting**
   365-day linear schedule for team allocation
3. **TimelockController**
   10 s delay (testnet); open executor; deployer as admin
4. **MyGovernor**
   Grants PROPOSER_ROLE to governor; EXECUTOR_ROLE to 0x0
5. **Treasury + Box**
   Both owned by Timelock; DAO-gated withdrawals
6. **GameToken×2 + AMM**
   HAR (Harmony) + SPK (Sparkle) pair; AMM deploys LPToken
7. **Equestria1155**
   VRF coordinator + keyHash + subId from env
8. **Factory + UUPS Proxy**
   GameTokenV1 impl → ERC1967Proxy → upgradeable to V2
9. **NFTRentalVault**
   GOV asset; PINKIE_PIE boosted NFT; owner = Timelock
10. **PriceFeedConsumer + VRF**
    ETH/USD Arb Sepolia; standalone VRF consumer

Frontend & Subgraph

Next.js 14 · wagmi v2 · Apollo Client · The Graph Studio

Subgraph (The Graph)

- schema.graphql + AssemblyScript mappings
- Indexes: Transfer, CraftRequested/Fulfilled, Swap
- NFTStaked/Unstaked, YieldInjected, VRF events
- Hosted on The Graph Studio → GraphQL endpoint

Pages (Next.js 14 App Router)

- /amm — swap & liquidity management
- /craft — element burn → pony minting UI
- /dao — propose, vote, queue, execute
- /inventory — ERC-1155 balance viewer
- /vault — stake NFT, deposit GOV, withdraw

Write Transactions (wagmi useWriteContract)

- craft(), swap(), addLiquidity(), removeLiquidity()
- stakeNFT(), unstakeNFT(), deposit(), withdraw()
- propose(), castVote(), queue(), execute()
- release() (vesting)

ABIs / Addresses (contracts.ts)

- ~216 KB auto-generated ABI + address map
- All 17 deployed contracts covered
- Imported by all page components
- Subgraph queries in dapp/src/lib/subgraph.ts

Security Findings & Recommendations

Manual review + Slither annotations + Foundry fuzz/invariant tests

Sev	Contract	Issue	Recommendation
	Equestria1155	VRF sub funding — elements burned, pony never minted	Document LINK requirement; consider state store for re-mint on retry
	GameTokenV1/V2	Upgrade owner = deployer EOA in current deploy script	Transfer proxy ownership to Timelock before mainnet deployment
MED	**PriceFeedConsumer**	10 min staleness fine for testnet, too short for ETH/USD mainnet	Use 3600 s (1 hr) for mainnet; document per-feed recommendations
MED	**AMM**	No MEV / sandwich protection	Frontend should default minAmountOut ≥ 99% of quoted output
LOW	**Equestria1155**	URI event not emitted (ERC-1155 spec requirement)	Emit URI event for each token ID on deploy or in constructor
INFO	**All contracts**	Minimal NatSpec / inline comments	Add @notice, @param, @return to all public/external functions

Summary

- 13 smart contracts — fully implemented and tested
- 80+ Foundry test cases: unit, fuzz, invariant, fork
- Chainlink VRF v2.5 + Price Feeds with all safety guards
- Full DAO: 1M GOV, linear vesting, Treasury, Box, Governor
- ERC-4626 vault, CREATE2 factory, UUPS proxy — all demonstrated
- Next.js dApp + The Graph subgraph for all core flows